The Invariance Principle — problem 1.31

A. Engel — Problem-Solving Strategies · 1998

Let $a_1, a_2, \ldots, a_n$ be a permutation of $1, 2, \ldots, n$. If n is odd, then the product $P = (a_1 - 1)(a_2 - 2)\ldots(a_n - n)$ is even. Prove this.